

Calendar

REAL VARIABLES I

Fall 2026

I will put all deadlines on your Blackboard Calendar. Please check it regularly!

0.1 Key university dates

For official University dates, see the Fall 2026 academic calendar and the Registrar's final-exam schedule.

Date	Event
Monday, August 17	First day of classes
Friday, August 21	Last day to add this course through the standard add period
Friday, August 28	Last day to drop without a W
Thursday, December 3	Last day of classes and last day to withdraw from all classes

0.2 Exams

Date	Event
Week of Wednesday, October 12 - Oct 18, 2026, TENTATIVE	Written midterm (two hours, outside the regular class)
Wednesday, December 9, 2026, 10:15 - 12:15	Final-exam period

0.3 Daily lecture log

Wk	Lec
	L1: Riemann integration and its limitations
2026-08-17	Review: the Riemann integral; theorem+proof: continuous functions are Riemann integrable; review: c
	L2: Countability and unbounded functions
2026-08-19	Proposition + proof: continuous functions with L1 norm are not a complete metric space. Review: exi
	L3: Countability, the extended reals, and sums of non-negative numbers
2026-08-21	Review: fat rationals; sketch: Riemann integrable functions with respect to L^1 are incomplete. Finite,
	L4: Sums of non-negative numbers
2026-08-22	Review: continuity of arbitrary non-negative sums with respect to increasing index sets Corollary: ever
	L5: Outer measures generated by covering costs
2026-08-26	Motivation: construct a notion of size from elementary sets whose costs are already known. Define a c
	L6: Lebesgue outer measure on $\mathbb{R}^d$ TENTATIVE
2026-08-28	Proposition+proof (maximality): μ^* is the largest outer measure whose value on every set in $\mathcal{A}$ is boun
	L7: Properties and examples of Lebesgue outer measure TENTATIVE
2026-08-30	Proposition+proof: translation invariance of λ^* , obtained by translating covers; statement and proof of
	L8: The Vitali obstruction TENTATIVE
2026-09-02	Complete the Vitali argument. Enumerate the relevant rational translates of a representative set, prov
	L9: Outer-measure problem session and transition to measurable sets TENTATIVE
2026-09-04	Compute the outer measure generated by weighted singleton costs and justify the formula as a non-neg
	L10: Algebras, σ-algebras, and generators TENTATIVE
2026-09-04 ⁴	Definitions: rings of sets, algebras, and σ -algebras; closure consequences from complements and De Mo
	L11: Measures and their basic inequalities TENTATIVE
2026-09-11	Definition: a measure on a σ -algebra and a measure space. Examples: counting measure, weighted cou
	L12: Continuity of measures and Borel sets TENTATIVE
2026-09-14	Define increasing and decreasing sequences of sets, together with set $\liminf$ and $\limsup$; verify that a c
	L13: Carathéodory measurability I: the criterion and finite operations TENTATIVE
2026-09-16	For an outer measure μ^* , define a Carathéodory measurable set A by the splitting identity $\mu^*(E) = \mu^*(E \cap A) + \mu^*(E \cap A^c)$
	L14: Carathéodory measurability II: the extension theorem TENTATIVE
2026-09-18	Theorem+proof: the Carathéodory measurable sets form a σ -algebra. First handle finite disjoint union
	L15: Lebesgue measurable sets and Lebesgue measure TENTATIVE
2026-09-20	Proposition+proof: coordinate half-spaces, and hence rectangles, satisfy the Carathéodory criterion for
	L16: Dynkin's π-λ theorem and uniqueness of measures TENTATIVE
2026-09-23	Define π -systems and Dynkin (λ)-systems and compare their closure properties with those of a σ -algeb
	L17: Measurable functions TENTATIVE
2026-09-25	Begin the next unit. Define a measurable map $f: (X, \mathcal{A}) \rightarrow (Y, \mathcal{B})$ by measurability of inverse images.
	TENTATIVE: Future lecture topics and their placement may change as the course develops.